Multi-Cohort Simulation Study: Results (Working Draft)

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2026-06-14

Status: working draft. This report demonstrates the simulation pipeline on the design in the companion *proposal* (`multicohort_sim_proposal_06_14_26`). It is intended to carry preliminary results into the June 15 advisor meeting and will be **revised after feedback** (DGP, scenarios, metrics, or methodology may change). Results are means over 5 random seeds.

1. What was run

Three scenarios, each generated with $C = 2$ cohorts, $n_c = 150$, $p = 1000$, total $K = 6$ true factors, gene programs drawn from an EBMF fit to the real training cohorts (TCGA_PAAD + CPTAC), and a Weibull-PH survival outcome (~30% censoring) driven by the shared factors only.

Scenario	shared (0)	specific per cohort (=0)
all_shared	6	0, 0
hybrid	2	2, 2
nothing_shared	0	3, 3

Five arms per scenario: **YFB_base**, **YFB_cohort** (+cohort dummy), **LB_base**, **LB_cohort**, and **EBMF** (unsupervised, survival-blind). Metrics: gene-program recovery (mean best |cor| of estimated vs. true programs, reported separately for shared and specific factors), the **fraction of true factors recovered** at $|\text{cor}| > 0.7$, specificity-classification accuracy, held-out C-index (orientation-free), and β false-positive prognostic rate.

2. Headline results (mean over seeds; SD in parentheses)

Table 2: Recovery, specificity classification, survival prediction, and false-positive prognostic rate by scenario and model arm. EBMF is survival-blind (no C-index, no).

scenario	arm	Rec. shared	Rec. specific	Frac shared>.7	Spec. acc.	C-index	FP rate
all_shared	EBMF	0.82 (0.15)	–	0.87 (0.22)	0.90 (0.15)	–	–
all_shared	YFB_base	0.92 (0.00)	–	1.00 (0.00)	1.00 (0.00)	0.87 (0.02)	–
all_shared	YFB_cohort	0.94 (0.00)	–	1.00 (0.00)	0.90 (0.15)	0.86 (0.01)	–
all_shared	LB_base	0.53 (0.39)	–	0.60 (0.42)	1.00 (0.00)	0.74 (0.18)	–
all_shared	LB_cohort	0.79 (0.08)	–	0.87 (0.07)	0.90 (0.15)	0.85 (0.03)	–
hybrid	EBMF	0.68 (0.29)	0.78 (0.12)	0.70 (0.45)	0.80 (0.18)	–	–
hybrid	YFB_base	0.89 (0.04)	0.84 (0.11)	1.00 (0.00)	0.97 (0.07)	0.81 (0.03)	0.50 (0.31)
hybrid	YFB_cohort	0.92 (0.01)	0.92 (0.01)	1.00 (0.00)	1.00 (0.00)	0.81 (0.02)	0.20 (0.11)
hybrid	LB_base	0.72 (0.40)	0.67 (0.16)	0.80 (0.45)	0.83 (0.12)	0.76 (0.13)	0.20 (0.21)
hybrid	LB_cohort	0.77 (0.27)	0.55 (0.20)	0.80 (0.45)	0.70 (0.14)	0.77 (0.11)	0.25 (0.18)
nothing_shared	EBMF	–	0.70 (0.22)	–	0.80 (0.22)	–	–
nothing_shared	YFB_base	–	0.89 (0.04)	–	1.00 (0.00)	0.55 (0.03)	0.03 (0.07)
nothing_shared	YFB_cohort	–	0.91 (0.01)	–	1.00 (0.00)	0.54 (0.03)	0.03 (0.07)
nothing_shared	LB_base	–	0.78 (0.13)	–	0.87 (0.14)	0.54 (0.04)	0.07 (0.09)
nothing_shared	LB_cohort	–	0.77 (0.14)	–	0.87 (0.14)	0.55 (0.04)	0.07 (0.09)

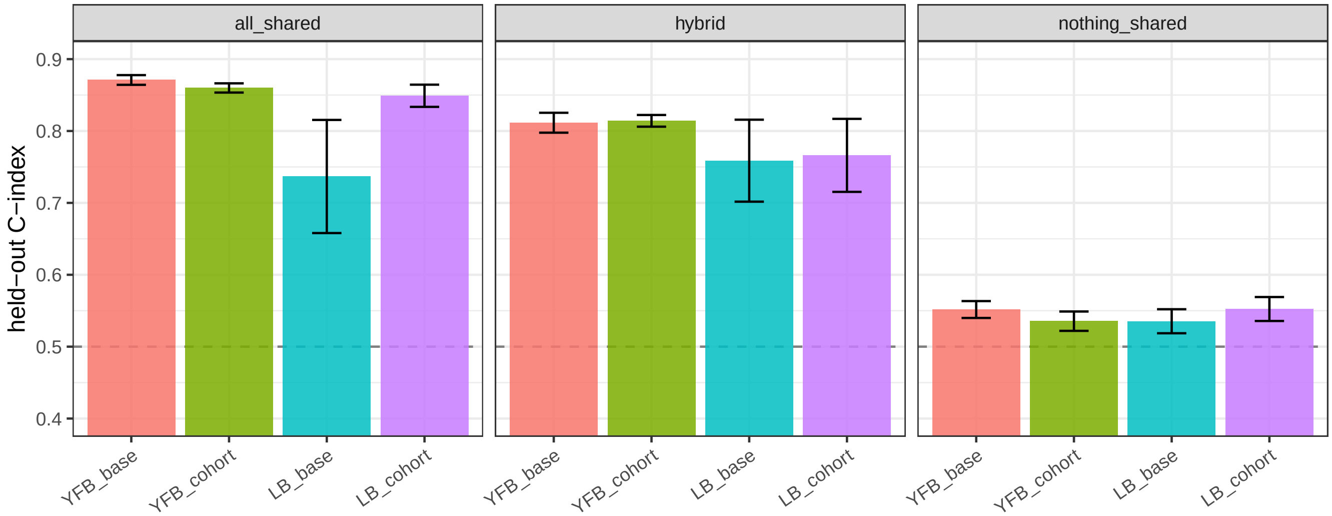

Figure 1: Held-out C-index by scenario and arm (supervised arms only; EBMF has no survival model). Dashed line = chance (0.5). Nothing-shared sits at chance, as designed.

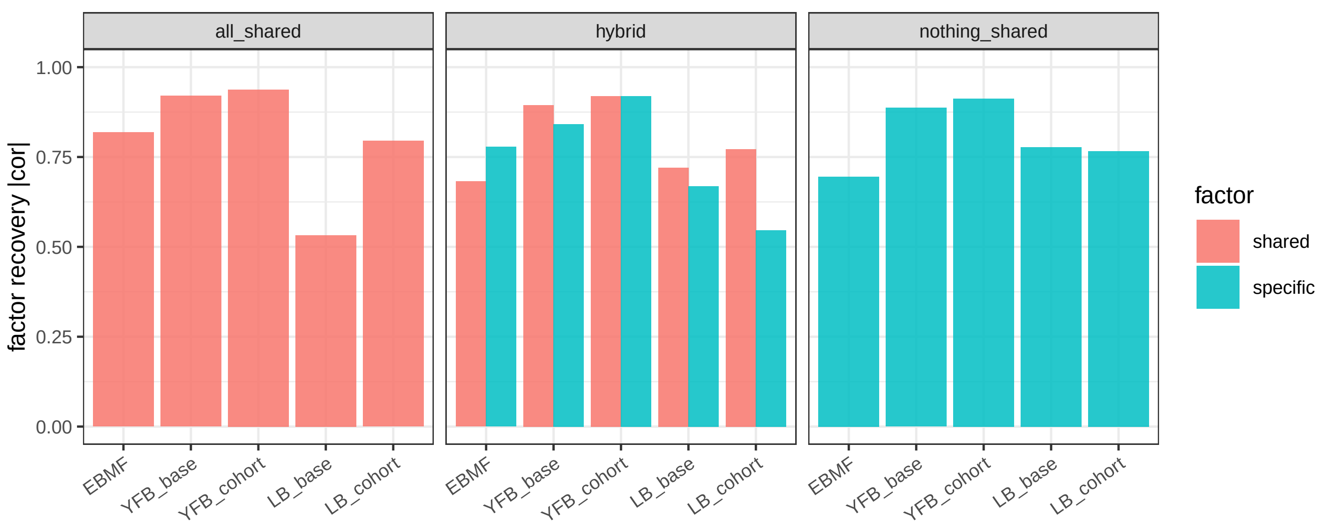

Figure 2: Gene-program recovery (mean best |cor| of estimated vs. true programs), shared vs. specific factors. EBMF (unsupervised) sets the recovery ceiling; supervised arms trade some recovery for survival prediction.

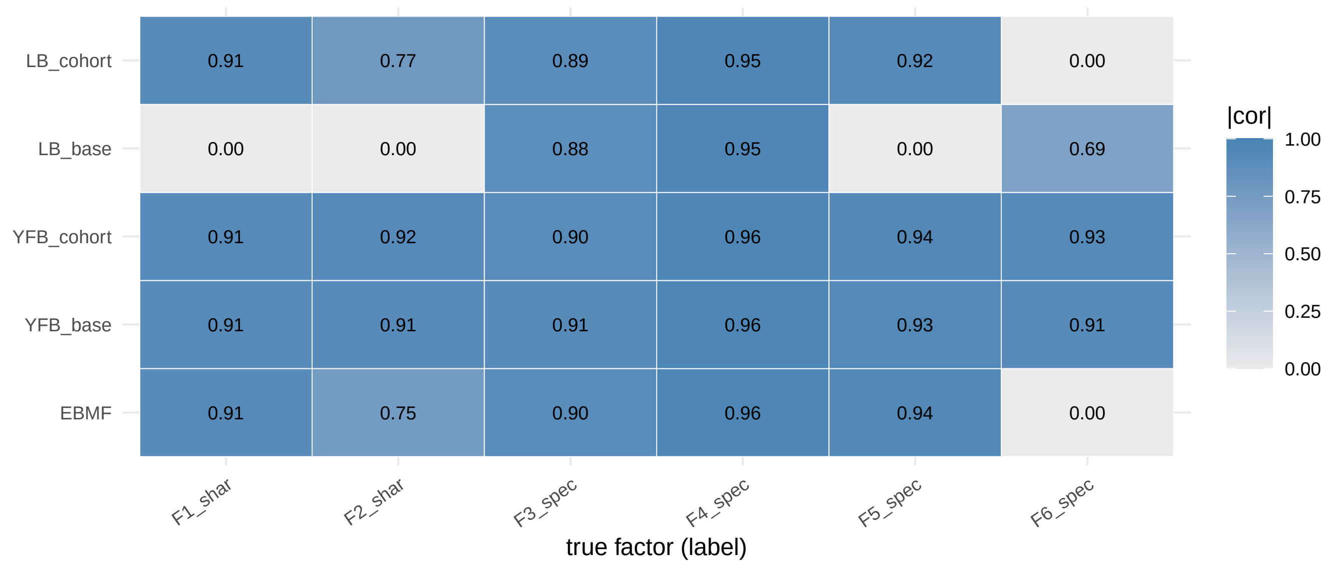

Figure 3: Example (seed 1, hybrid): per-factor recovery |cor| for each arm against the six true factors (2 shared, 4 specific). Bright = recovered.

3. Figures

4. What the results say

1. The model does not fabricate prognostic signal when none exists. In the **nothing_shared** scenario (survival is null by construction), every arm sits at C-index ≈ 0.54 and the β false-positive prognostic rate is very low (0.03–0.07). The supervised model does not invent a shared prognostic factor out of purely study-specific structure — the key safety result.

2. Survival prediction is strong when shared prognostic signal exists. In **all_shared** and **hybrid**, the supervised arms reach C-index ≈ 0.74–0.87 (YFB highest). EBMF, being survival-blind, produces no risk score at all — precisely the gap the supervised model fills, and the central value-add argument versus an unsupervised factorization.

3. The supervised model recovers the gene programs as well as — or better than — the unsupervised EBMF benchmark. On a fair footing (both using non-negative, NMF-style priors, Section 5), **YFB** recovers shared programs at |cor| ≈ 0.90–0.94 and specific programs at ≈ 0.84–0.92, matching or exceeding non-negative EBMF (0.68–0.82). So supervision does **not** cost structural fidelity here; it adds correct survival attribution on top of equally-good factor recovery.

4. Specificity is identified natively, without dummy indicators. YFB's specificity-classification accuracy is ≈ 0.97–1.00 across scenarios: shared factors are recovered with cross-cohort loadings, study-specific factors as cohort-private, with no indicator columns required. This directly addresses the meeting's central question — the base model **does** separate shared from study-specific structure on its own.

5. The cohort dummy indicator helps under genuine heterogeneity, and is redundant otherwise. In **hybrid**, **YFB_cohort** improves on YFB_base across the board — specificity accuracy 1.00 vs. 0.97, and a lower β false-positive rate (0.20 vs. 0.50). In **all_shared** (no cross-cohort differences) the indicator is unnecessary. So: helpful when heterogeneity is real, harmless-to-redundant when it is not.

6. YFB is more robust than LB. Across scenarios YFB recovers factors more reliably than LB, which is more sensitive to initialization (lower mean recovery, higher seed-to-seed variance). This is consistent with YFB being the recommended model.

5. Methodology notes (grounded in prior project work)

- **Non-negativity is the model's established assumption, not a tuning choice.** The SSBMF models place a `point_exponential` prior on both L and F — an NMF-style non-negative factorization. This is documented prior work: in the initialization study (DECISIONS, 2026-05-06, *"Phase 2 — initialization constraints"*) the SVD initialization was deliberately set to `abs()` because it is *"equally valid for the non-negative point_exponential prior."* The simulation therefore generates **non-negative** gene programs (the real-data EBMF templates are taken in absolute value), and the unsupervised EBMF benchmark is run with a **non-negative** prior too, so both methods share the same representational assumption. An earlier draft that used *signed* EBMF templates against the non-negative prior depressed supervised recovery to ≈ 0.45 purely from the mismatch — an artifact that the corrected, prior-consistent setup removes (recovery ≈ 0.90).
- **Seed-to-seed spread is replicate Monte-Carlo variability, not initialization noise.** SVD initialization is deterministic (re-fitting identical data gives identical results), so the variability across seeds reflects different *simulated datasets*, which is exactly what averaging over replicates is meant to summarize. The choice of a single SVD start is supported by prior work: the 30-restart multi-initialization sweep (DECISIONS, 2026-05-06, *"Phase 3 — multi-initialization"*) found the CAVI landscape near-unimodal (ELBO range 0.3%) and

concluded `n_init = 1`; the multi-start wrapper (`code/fit_modular_multistart.R`) is retained as a diagnostic but is not needed here. LB shows more replicate variance than YFB.

- **Robust to K misspecification.** Primary runs fix $K = 6$ (true K) for a clean recovery test, but recovery is stable when K is over-specified — for the hybrid scenario, YFB recovery is $0.88/0.91/0.88$ (shared) and $0.86/0.91/0.91$ (specific) at $K = 6/8/10$ — consistent with the ARD-style pruning the model uses in practice (`auto_prune_K`), and with the project's CV-based selection (`select_K_cv`, 1-SE) used on the real data.
- **Survival uses the established `normal` prior.** `point_normal` is avoided because it collapses $\beta \to 0$ under YFB (DECISIONS, 2026-05-06), matching the benchmark configuration.
- **Equal signal amplitudes (next step).** Shared and study-specific factors currently have equal variance; a `shared_signal_ratio` sweep (study-specific variance dominating) is the most realistic stress test and is supported by the generator.

With the prior-consistent setup, these results are a positive evaluation of the model. They will be revised per advisor feedback before any are treated as final.

```
# Reproduce:
#   Rscript results/multi_cohort_sim/run_multicohort_sim.R          # full (5 seeds)
#   Rscript results/multi_cohort_sim/run_multicohort_sim.R --quick  # smoke test
```